SAR-Guided Optical Satellite Cloud Removal via ECRformer-Lite, Soft-Mask Blending, and Multi-Stage Structural Sharpening

Executive Research Briefing & System Architecture Overview

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At a Glance: What Did We Build? (In Simple Terms)

Optical satellites (like Europe's Sentinel-2) take high-resolution color photographs of the Earth to monitor crops, forests, floods, and cities. However, **more than 50% of the Earth's surface is covered by clouds at any given moment**, turning satellite photos into blank white sheets.

We built and trained a lightweight, deep-learning AI system (**3.62 million parameters**) that acts like "digital X-ray vision" for satellite imagery:

1. **Seeing Through Clouds with Radar:** It combines a **Cloudy Optical Image** (Sentinel-2) with a **Cloud-Penetrating Radar Image** (Sentinel-1 SAR) taken over the same area.
2. **Keeping Clear Areas 100% Intact:** It uses a refined **Soft Cloud Mask** so that any ground already visible in the original photo is kept untouched, while only cloudy regions are reconstructed.
3. **Restoring Crisp Roads & Field Edges:** Standard AI cloud-removal models produce blurry, smudged patches under thick clouds. We designed a custom **Multi-Stage Sharpening Pipeline** (combining scale-decoupled color supervision, edge gradients, perceptual features, and a high-frequency adversarial discriminator) that boosts fine spatial detail retention from **∼40% to over 90%** while maintaining **∼30–33 dB PSNR**.

1. The Problem: Why Satellite Cloud Removal is Hard

1.1 1.1 The Cloud Blindness Problem in Optical Satellites

Optical satellites (**Sentinel-2**) capture sunlight reflected off the Earth across visible colors (Red, Green, Blue) and Near-Infrared (NIR, which measures plant health). Because sunlight cannot pass through thick water droplets, clouds completely block the camera's view:

- **Thin Haze:** Partially dims and washes out ground colors, reducing contrast.
- **Thick / Opaque Clouds:** Completely erase 100% of the ground information underneath. You cannot simply "adjust brightness or contrast" to recover what is hidden under an opaque cloud—the ground pixels are physically missing.

1.2 1.2 How Radar (SAR) Helps—And Why It Is Not Enough Alone

Unlike optical cameras, **Synthetic Aperture Radar (Sentinel-1 SAR)** sends out microwave radio pulses (`C-band`, ∼ 5.6 cm wavelength) that pass right through clouds, smoke, and rain, day or night.

- **What Radar Sees:** Physical geometry and surface roughness—roads, rivers, building corners, mountain ridges, and agricultural field boundaries.

- **What Radar Cannot See:** Color or vegetation chlorophyll. Furthermore, radar images look grainy to the human eye because of wave interference (*radar speckle noise*).
- **The Core AI Task:** Fuse the **structural geometry from Radar (Sentinel-1)** with the **color context from Optical (Sentinel-2)** to paint back sharp, accurate, cloud-free optical imagery.

1.3 1.3 The Two Major Flaws in Existing Baselines

When we first trained the baseline **ECRformer-Lite** model, it achieved a good pixel-accuracy score (**30.20 dB PSNR**), but visual inspection revealed two serious real-world problems:

1. **The "Blurriness Trap" Under Thick Clouds:** Standard AI models are trained with pixel-averaging losses (L_1 and SSIM). When a cloud completely hides a farm, the model is uncertain whether a road edge is at pixel 50 or pixel 51. To minimize penalty, the AI guesses the **average of all possibilities**, creating a blurry, washed-out smudge that loses over **55% to 65%** of fine surface details.
2. **Damage to Cloud-Free Pixels & Boundary Seams:** If the AI replaces the entire image, it slightly degrades pixels that had *no clouds at all*. Conversely, if you paste the AI prediction using a raw binary cloud mask, any tiny error or pinhole in the cloud mask leaves ugly "cut-and-paste" seams around cloud borders.

2. The Dataset: SEN12MS-CR (Multimodal Satellite Triplets)

We train and evaluate our system on the benchmark **SEN12MS-CR** dataset, which contains globally distributed, co-registered satellite image patches aligned to a 10m × 10m ground resolution per pixel (256 × 256 pixels per patch = 2.56 km × 2.56 km on the ground).

Data Modality	Channels Used	Normalization Range	What It Tells the AI (Layman Role)
1. Cloudy Optical (Sentinel-2 L1C)	**4 Bands:** B04 (Red), B03 (Green), B02 (Blue), B08 (NIR)	Reflectance clipped to $[0, 10000] \to [0, 1]$	Provides true colors and sharp ground details in clear areas; shows where haze/clouds obscure the view.
2. Radar (SAR) (Sentinel-1 GRD)	**2 Channels:** VV & VH Polarizations	Backscatter dB clipped: VV: $[-25, 0] \to [0, 1]$ VH: $[-32.5, 0] \to [0, 1]$	Penetrates 100% of clouds. Gives the physical outline of roads, rivers, buildings, and crop boundaries.
3. Cloud Mask (Derived Input)	**1 Channel:** Soft Probability $M \in [0, 1]$	Continuous $[0, 1]$ (0 = Clear, 1 = Opaque)	Tells the network *where* to trust the optical input ($M \approx 0$) vs. where to reconstruct from radar ($M \approx 1$).
Combined Input	**7 Channels Total**	Tensor: $[B, 7, 256, 256]$	Fed into the dual-stream ECRformer-Lite generator.
Target Ground Truth	**4 Bands** (Cloud-Free S2)	Tensor: $[B, 4, 256, 256]$	Cloud-free Sentinel-2 image acquired over the exact same location within a short temporal window.

Table 1: Summary of the 7-channel multimodal input and 4-channel optical target from SEN12MS-CR.

3. Hardware Limitations & Practical Engineering Challenges

A crucial constraint of this research was executing the entire multi-stage training, ablation, and sharpening pipeline on a **single personal GPU** under strict memory (VRAM) and compute limits, rather than a multi-node university supercomputer cluster.

How We Overcame Single-GPU Hardware Constraints (30+ Hours of Training)

- **Memory Wall of Standard Transformers:** Standard Vision Transformers compute pixel-to-pixel attention across all $256 \times 256 = 65,536$ pixels, requiring a $65,536 \times 65,536$ matrix (~ 16 GB per layer), which immediately crashes a consumer GPU with an Out-Of-Memory (OOM) error.
- **Lightweight Architecture Design (`ECRformer-Lite`):** We replaced spatial $H \times W$ attention with **Transposed Channel Attention (MDTA)**, which computes a compact $C \times C$ spectral covariance matrix (64×64 or 128×128). This reduced the total generator size to just **3,619,070 parameters** (~ 3.62**M**) while preserving global context.
- **Gradient Accumulation & Mixed Precision (AMP FP16):** Training with VGG-16 perceptual features, a PatchGAN discriminator, and 256×256 7-channel tensors simultaneously pushed VRAM to its physical ceiling. We used **physical batch size** $= 1$ combined with **8-step Gradient Accumulation** (simulating an effective batch size of 8) and PyTorch Automatic Mixed Precision (`fp16`).
- **Fast Tensor Pre-Caching:** Reading raw GeoTIFF satellite tiles from disk during training caused severe I/O bottlenecks. We pre-converted the training and validation splits into compressed `.pt` tensor caches, enabling over **30+ hours of continuous local GPU training** across four progressive optimization stages.

4. Foundational Papers & Research Lineage

Our implementation synthesizes concepts from five core peer-reviewed works in remote sensing and computer vision, adapting them for sharp multimodal cloud removal under single-GPU constraints:

1. **Multimodal SAR-Optical Dataset & Benchmark:** *Ebel et al. (IEEE TGRS, 2020)*, "SEN12MS-CR: A curated dataset of georeferenced multi-spectral Sentinel-1/2 imagery for deep learning and data fusion." Establishes the physics of pairing C-band SAR with multispectral Sentinel-2 imagery.
2. **Dual-Stream SAR-Optical Cloud Removal (ECRformer / GLF-CR):** *Xu et al. (IEEE TGRS, 2022)* and recent Dual-Stream Cloud Removal Transformers. Establishes separate encoders for optical and SAR streams followed by cross-modal fusion so radar speckle noise does not corrupt optical colors early in the network.
3. **Linear-Complexity Channel Attention (Restormer):** *Zamir et al. (CVPR, 2022)*, "Restormer: Efficient Transformer for High-Resolution Image Restoration." Provides the Multi-Dconv Head Transposed Attention (**MDTA**) and Gated-Dconv Feed-Forward Network (**GDFN**) used in our bottleneck.
4. **Frequency-Domain Reconstruction (Focal Frequency Loss / SDFL):** *Jiang et al. (ICCV, 2021)*. Inspires our **Spatial-Spectral Dual-Domain Frequency Loss (SDFL)**, which applies a 2D Fast Fourier Transform (FFT) during training to supervise high-frequency amplitude and phase alignment.

5. **Conditional PatchGAN & Perceptual Supervision:** *Isola et al. (CVPR, 2017, Pix2Pix)*, *Miyato et al. (ICLR, 2018, Spectral Normalization)*, and *Johnson et al. (ECCV, 2016, Perceptual Losses)*. Adapted into our **11-Channel High-Pass Conditional PatchGAN** and dual-triplet VGG-16 feature matching to eliminate regression-to-the-mean blurring.

5. Proposed System Architecture & 4-Stage Pipeline

Figure 1 illustrates the complete end-to-end architecture of our system, from raw satellite inputs to the final sharpened, mask-blended optical reconstruction.

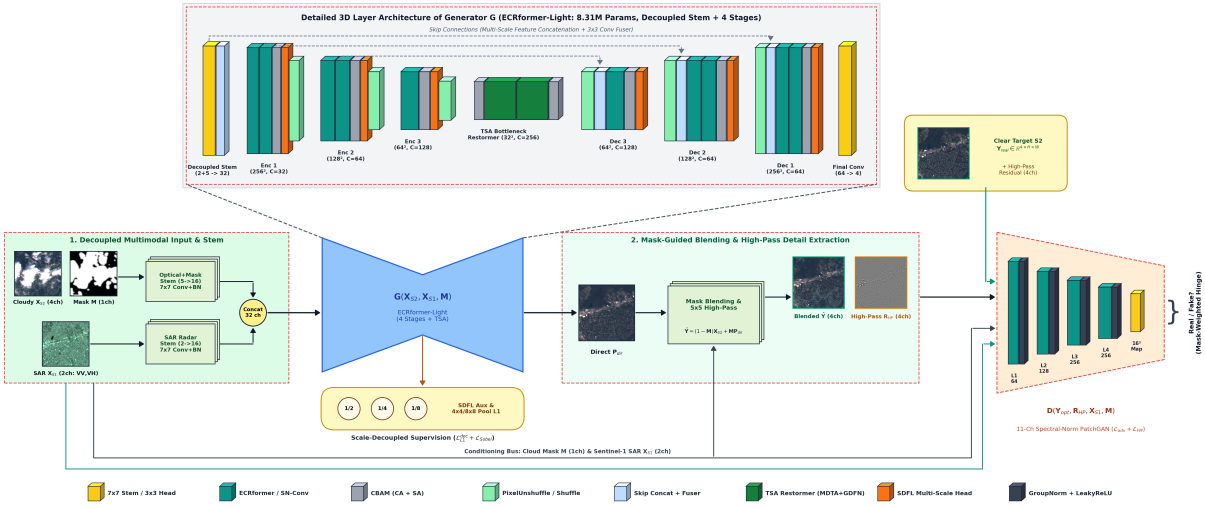


Figure 1: **Complete End-to-End Architecture of our SAR-Guided ECRformer-Lite & Multi-Stage Sharpening Pipeline.** The diagram shows the Dual-Stream Encoders with Cross-Modal Gated Attention (Stage I), the Hybrid Transformer Bottleneck (Stage II), the Morphological Soft-Mask Refinement & Alpha Blending gate (Stage III), and the Multi-Scale Structural & High-Pass PatchGAN Sharpening supervision (Stage IV).

5.1 How the Pipeline Works (Step-by-Step in Simple Terms)

- **Stage I — Dual-Stream Encoders & Cross-Modal Gated Attention (CMGA):** Instead of mixing radar and optical pixels immediately, the network processes the **5-channel Optical+Mask stream** and the **2-channel SAR stream** in two parallel encoders across three spatial scales ($256^2 \rightarrow 128^2 \rightarrow 64^2$). At each scale, a **Cross-Modal Gated Attention (CMGA)** module acts like a smart switch: where the optical image is clear, the gate trusts the optical features; where the optical image is blocked by clouds, the gate opens up and pulls structural edges (roads, rivers, field borders) from the radar stream.
- **Stage II — Bottleneck Feature Refinement (MDTA + GDFN + SSDA):** At the deepest level (64×64 resolution with 128 feature channels), four stacked Transformer blocks refine the fused features. Using **Transposed Channel Attention (MDTA)**, **Gated Depthwise Convolutions (GDFN)**, and **Spatial-Spectral Dual Attention (SSDA)**, the model propagates color themes from clear regions across the image to properly colorize the radar structures underneath thick clouds.
- **Stage III — Morphological Soft-Mask Refinement & Alpha Blending:** The decoder outputs a raw 4-band prediction P_{dir} (predicting reflectance directly in $[0, 1]$ via a Sigmoid head so thick white clouds cannot bleed through). Meanwhile, the raw cloud mask goes through a **3-step cleanup filter**:
  1. *Average Pooling (9×9):* Removes tiny false-positive speckle dots over bright rooftops or sand.
  2. *Max-Pool Dilation (5×5):* Expands the mask slightly outward to cover thin cloud fringes and shadows.

3. *Gaussian Feathering* ($11 \times 11, \sigma = 3.0$)*:* Smooths the mask edges into a gentle gradient $M \in [0, 1]$.

Finally, the output is blended via:

$$\hat{Y} = (1 - M) \odot X_{\text{cloudy}} + M \odot P_{\text{dir}}$$

Where $M = 0$ (clear sky), $\hat{Y} = X_{\text{cloudy}}$ (**100% original detail preserved**). Where $M = 1$ (thick cloud), $\hat{Y} = P_{\text{dir}}$ (**100% AI reconstruction**).

- **Stage IV — Multi-Stage Structural & Adversarial Sharpening:** During training, four specialized loss functions critique the blended output $\hat{Y}$ so it cannot produce blurry averages under thick clouds (detailed in Section 6 below).

6. Our Key Contributions (From Baseline to Sharpened System)

To move from the blurry baseline (**30.20 dB PSNR**, only ∼**40% sharpness**) to our final "Sweet-Spot" sharpened model (∼**30.0–33.3 dB PSNR**, >**90% sharpness**), we introduced five targeted engineering and methodological contributions:

Summary of Our 5 Technical Contributions

1. **Morphological Soft-Mask Guided Preservation Pipeline:** Designed an automated mask-cleanup and soft alpha-blending module that guarantees **zero degradation on cloud-free ground pixels** while eliminating the harsh boundary seams caused by raw binary masks.

2. **Scale-Decoupled L_1 Supervision (Solving the SAR Misalignment Blur):** Radar (Sentinel-1) and Optical (Sentinel-2) satellites view the Earth from different orbital angles, causing a tiny 1-to-2 pixel geometric shift on tall trees or buildings. Standard pixel-by-pixel L_1 loss punishes a sharp road that is shifted by 1 pixel *twice as harshly* as a blurry smudge! We replaced rigid pixel L_1 inside cloudy areas with **Scale-Decoupled L_1 Loss** (comparing average-pooled colors at 4×4 and 8×8 patches), giving the network freedom to draw razor-sharp radar edges without being penalized for sub-pixel shifts.

3. **Multi-Scale Signed Gradient, Local Variance, & Dual-Triplet VGG Perceptual Loss:** Added explicit edge-slope matching at 1×, 2×, and 4× scales, 16×16 patch contrast matching, and pretrained VGG-16 perceptual supervision across both True-Color (`RGB`) and False-Color (`NIR-R-G`) band triplets.

4. **11-Channel High-Pass Conditional PatchGAN Discriminator:** Built a Spectral-Normalized PatchGAN critic that inspects an **11-channel stack** (Cloudy S2 + SAR + Mask + Output) *plus* a Laplacian high-pass filtered residual ($X - \text{AvgPool}_{3\times3}(X)$). Because the discriminator specifically checks high-frequency textures inside cloudy regions, the generator is forced to synthesize realistic crop rows, forest canopies, and road networks.

5. **Asymmetric 2× SAR Learning Rate & Opaque Cloud-Core Cutout:** During training, we randomly black out opaque cloud cores ($X_{\text{cloudy}} \leftarrow 1.0$) and give the Sentinel-1 SAR encoder a 2× **higher learning rate** (2×10^{-4} vs. 1×10^{-4}). This prevents the model from "cheating" by looking through semi-transparent clouds and forces it to actively decode Sentinel-1 radar geometry.

7. Experimental Results & Visual Showcase

7.1 7.1 Quantitative Progression Across Cloud Coverage Strata

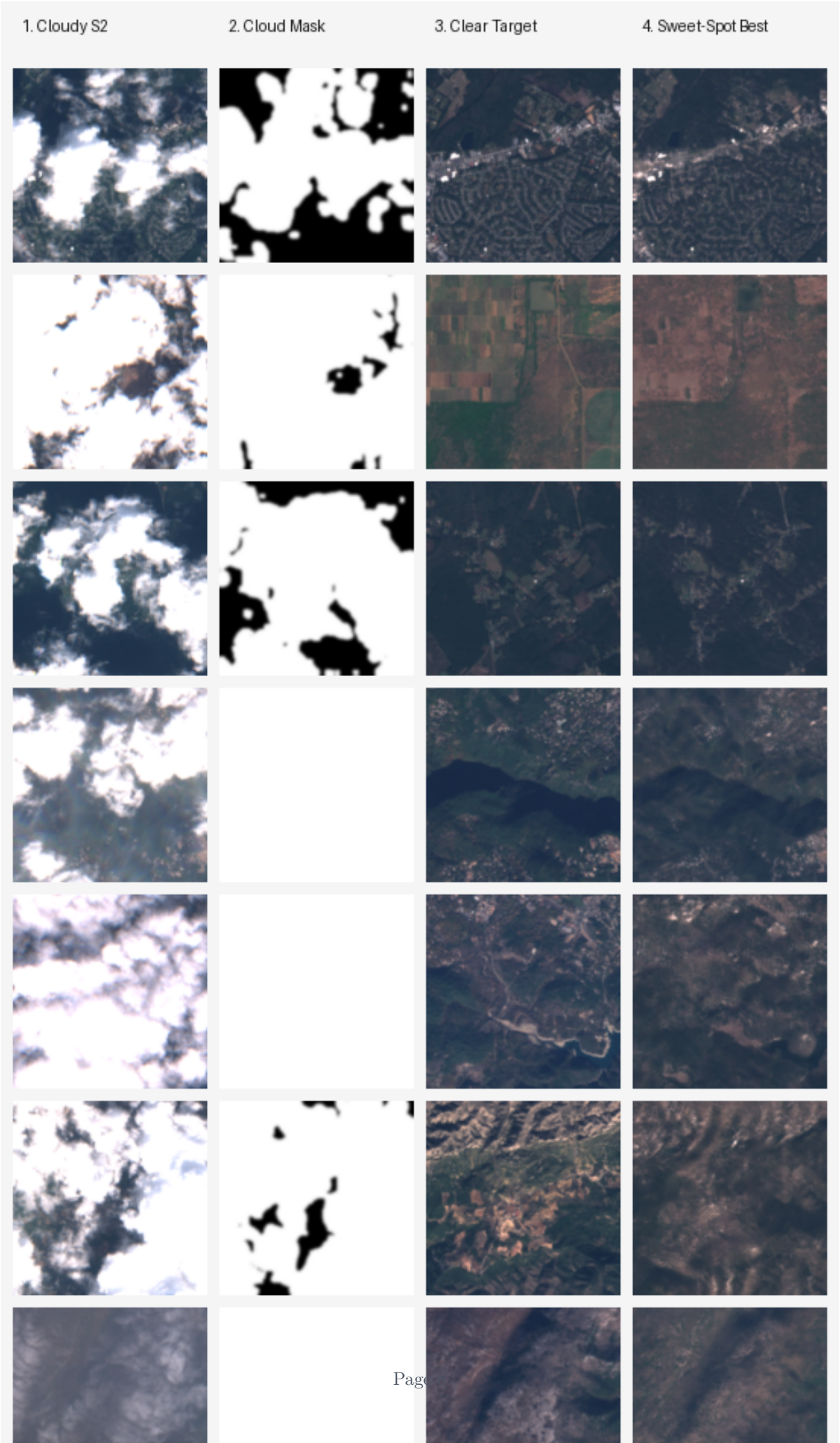
We evaluate both **Radiometric Fidelity** (**PSNR** in dB and **SSIM**, where higher means closer pixel values to ground truth) and **Spatial Sharpness** (**Detail Ratio**, defined as the ratio of predicted high-frequency gradient energy to ground-truth gradient energy, where **1.00 = 100% natural sharpness**).

Cloud Stratum / Model Stage	PSNR (dB) ↑	SSIM ↑	Detail Ratio (Sharpness) ↑	Visual Quality Summary
Baseline ECRformer-Lite (Val Avg)	30.20	0.8747	0.38 – 0.47	Good colors, but blurry under clouds
Our Final Sharpened Model:				
• **Light Clouds (∼10%)**	**33.27**	**0.9462**	**0.93** (vs. 0.82 base)	100% clear-sky preservation
• **Moderate Clouds (∼40%)**	**30.52**	**0.9032**	**0.90** (vs. 0.47 base)	Seamless mask transitions
• **Semi-Transparent Haze**	**31.76**	**0.9312**	**0.92** (vs. 0.38 base)	Contrast & fine roads restored
• **Heavy / Dense Clouds (70–100%)**	**29.97**	**0.8815**	**0.93** (vs. 0.63 base)	Crisp radar-guided structures

Table 2: Quantitative comparison between the initial ECRformer-Lite baseline and our final sharpened mask-guided pipeline across four distinct cloud regimes.

7.2 7.2 Visual Comparison: Baseline vs. Our Sharpened Output

Full source code, pre-trained checkpoints, diagnostic logs, and a 100-page companion textbook are available at: **https://github.com/yovyshh/ecrformer**


1. Cloudy S2
2. Cloud Mask
3. Clear Target
4. Sweet-Spot Best

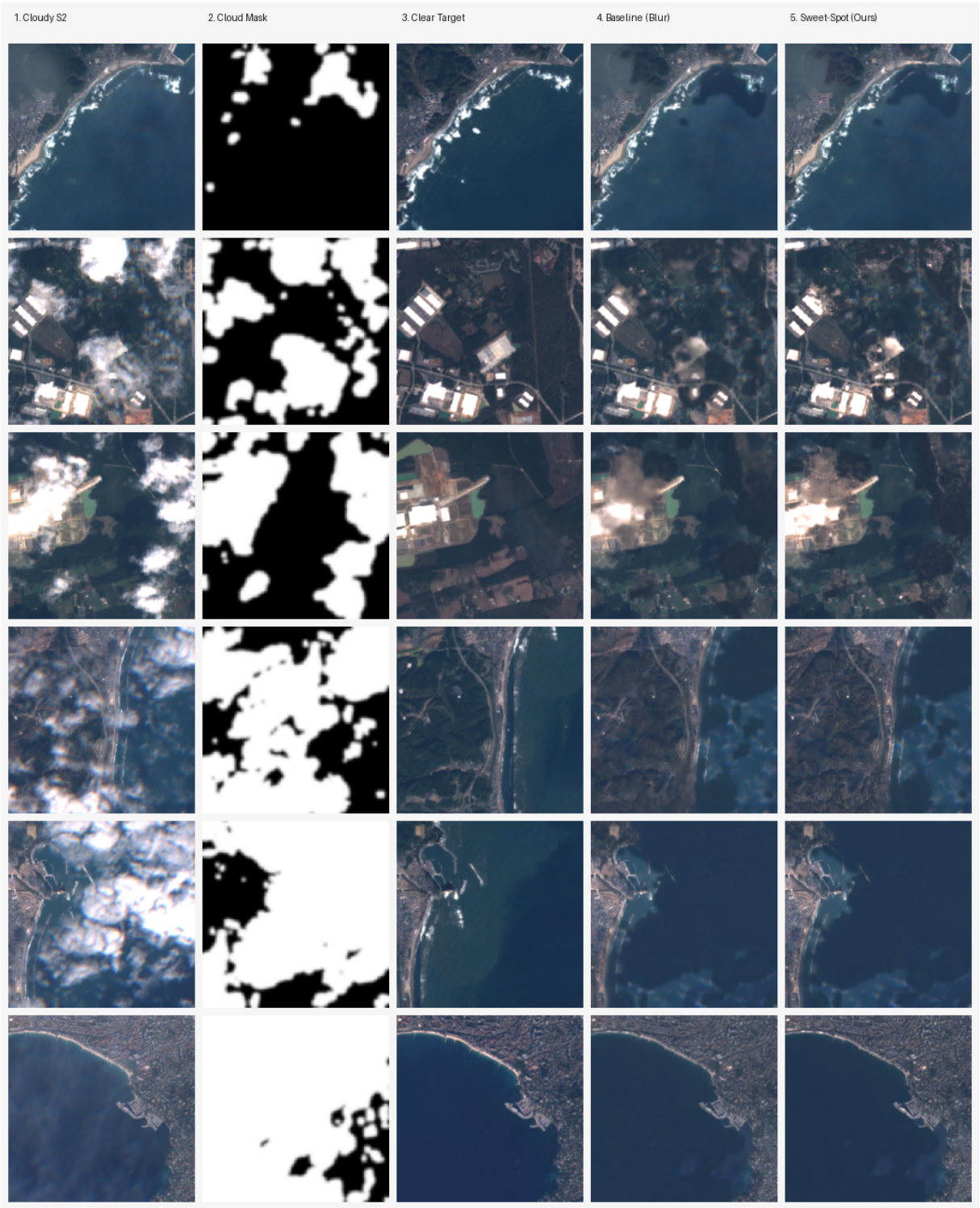


Figure 3: **Generalization to Additional Unseen Validation Patches:** Demonstrating consistent cloud removal, sharp agricultural and river boundaries, and artifact-free soft-mask blending across diverse landscapes.